Coding Theory and Mathematics

– Linear, Hamming, and Cyclic Codes –

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[기대수 3기 Colloquium]

수학의 즐거움, Enjoying Math

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I. Introduction

I. Introduction

Errors Are Everywhere

Physical world

- Scratches on a CD or DVD

Digital world

- Bit flips during internet transmission

Questions.

"How do we transmit (or store) data **reliably** in the presence of noise?"

I. Introduction

Can We Make Noisy Data Reliable Again?

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1 \cdots$$

$$\Big\downarrow \ \text{noise}$$

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 1 \cdots$$

I. Introduction

Can We Make Noisy Data Reliable Again?

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1 \cdots$$

noise

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ \textcolor{red}{0}\ 1\ 0\ 0\ 1 \cdots$$

- Can we **detect** it?

I. Introduction

Can We Make Noisy Data Reliable Again?

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1 \cdots$$

$$\Big\downarrow \text{ noise}$$

$$\cdots 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ {\color{red}0}\ 1\ 0\ 0\ 1 \cdots$$

- Can we **detect** it?
- Can we **fix** it?

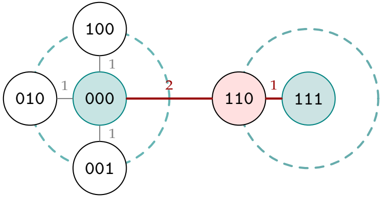
I. Introduction

Triple Repetition Code $[3,1]_2$

The simplest way to correct errors: **say it three times.**

$$0 \longmapsto 000, \qquad 1 \longmapsto 111.$$

Received	Majority	Decision
000	0×3	0
001	0×2	0
010	0×2	0
100	0×2	0
011	1×2	1
101	1×2	1
110	1×2	1
111	1×3	1



2-bit errors are detectable, but not always correctable.

Performance

- Detects any two-bit flips
- Corrects any single-bit flip

Cost

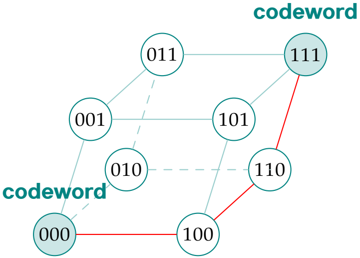
- Rate $R = 1/3$: send 3 bits to convey 1 bit
- Very inefficient!

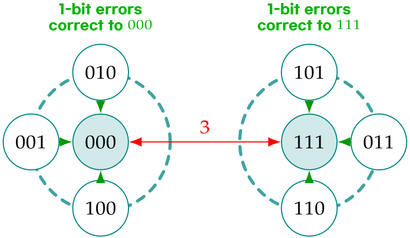
Question. Can we do much better?

I. Introduction

The Key Idea: Distance

Why does the triple repetition code work? The two codewords are **far apart**.





Minimum distance

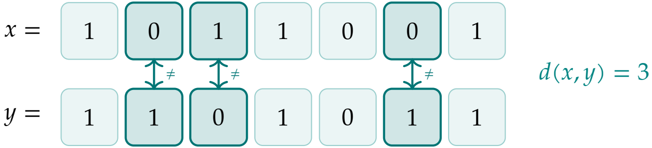
A code with minimum distance d corrects all errors of weight $\leq \left\lfloor \frac{(d-1)}{2} \right\rfloor$.

I. Introduction
Hamming Distance

Hamming distance counts how many coordinates differ. For any $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$,

$$d_H(\mathbf{x}, \mathbf{y}) = w_H(\mathbf{x} \oplus \mathbf{y}) = \#\{i : \; x_i \neq y_i\}$$

where w_H is Hamming weight (number of ones).

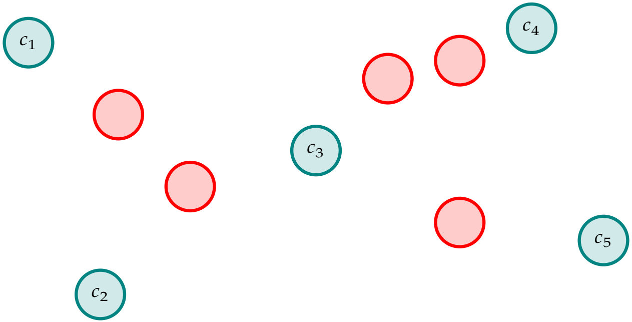


Example 1. $d_H(0101, 1100) = 2.$

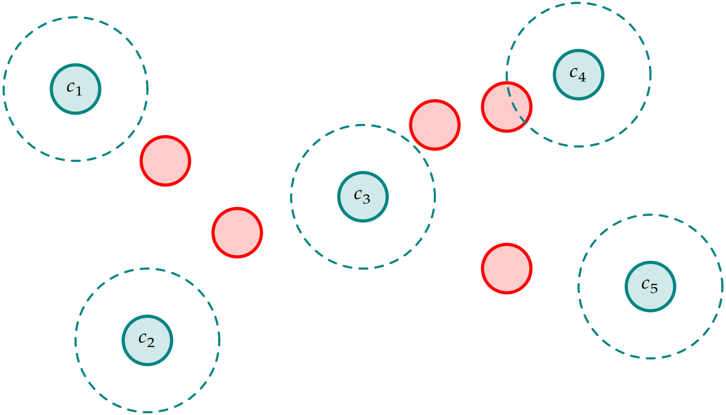
Example 2. $d_H(000, 101) = 2.$

Hamming distance is a metric.

I. Introduction
Correction Balls



I. Introduction
Correction Balls



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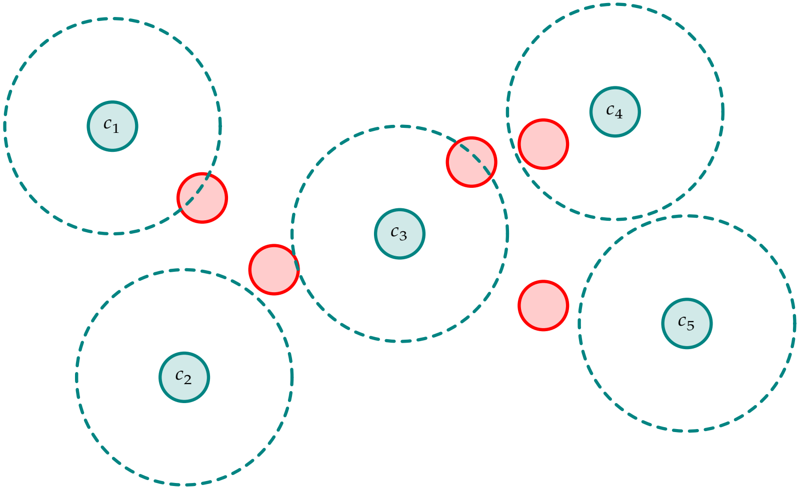
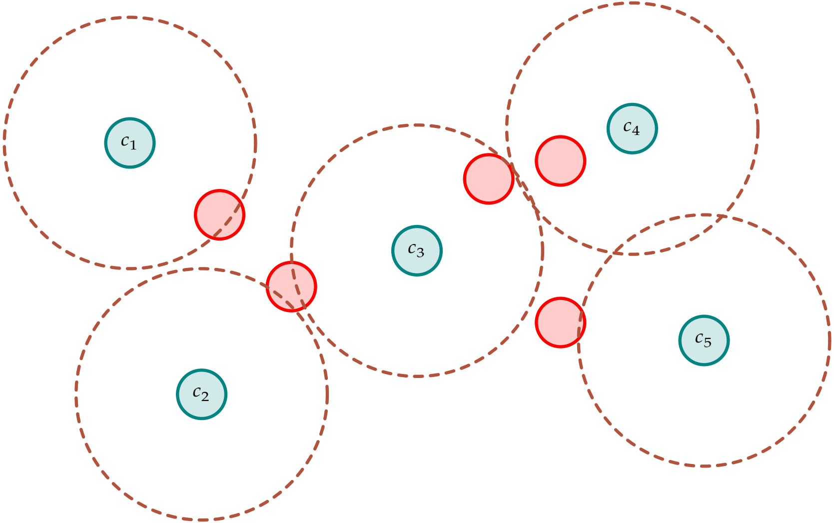


I. Introduction

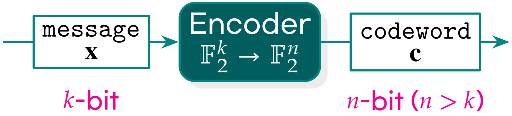
Correction Balls



Remark. **No overlap of correction balls** $\Longleftrightarrow$ unique decoding is possible.

I. Introduction

The Communication Model



Setup

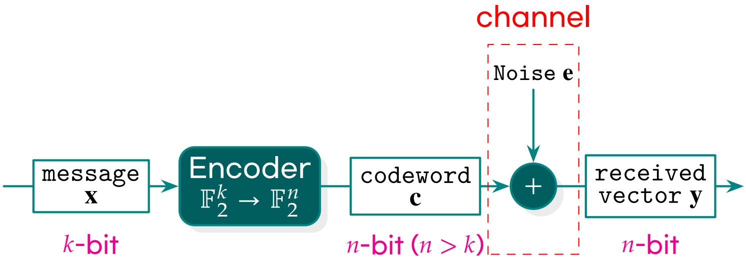
- Message $\mathbf{x} \in \mathbb{F}_q^k$
- Encoder : $\mathbf{x} \mapsto \mathbf{c} \in \mathbb{F}_q^n, \;\; n > k$
- Channel : $\mathbf{y} = \mathbf{c} \oplus \mathbf{e}$

Goal

- Recover $\hat{\mathbf{x}} = \mathbf{x}$ even when $\mathbf{e} \neq \mathbf{0}$
- Keep rate $R = k/n$ as large as possible

I. Introduction

The Communication Model



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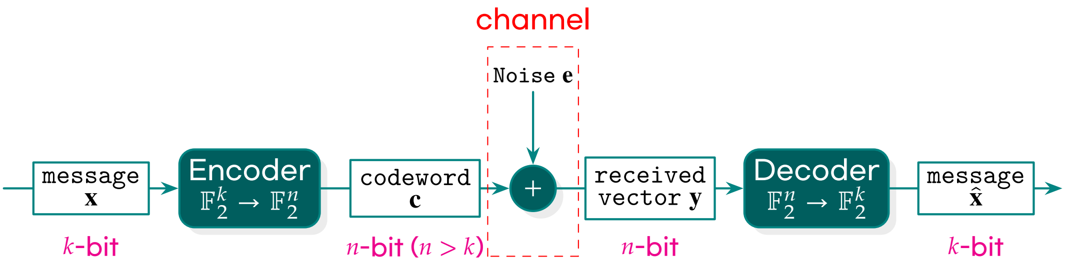

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Goal

- Recover $\hat{\mathbf{x}} = \mathbf{x}$ even when $\mathbf{e} \neq \mathbf{0}$
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I. Introduction

The Fundamental Tradeoff

Code	Rate	Detects	Corrects t
$[n,k,d]_q$-code	k/n	$d-1$	$\lfloor (d-1)/2 \rfloor$
Repetition $[3,1,3]_2$	$1/3 \approx 0.33$	2	1
Hamming $[7,4,3]_2$	$4/7 \approx 0.57$	2	1
BCH $[15,7,5]_2$	$7/15 \approx 0.47$	4	2
Reed-Solomon $[15,11,5]_{16}$	$11/15 \approx 0.73$	4	2
Reed-Solomon $[255,223,33]_{256}$	$223/255 \approx 0.87$	32	16

- high rate $\implies$ efficiency
- correct many errors $\implies$ powerful

I. Introduction
Roadmap

II. Linear Codes

Structure via **linear algebra**

- Subspaces of $\mathbb{F}_q^n$
- Generator Matrix G, parity-check H
- Hamming distance & error correction

IV. Cyclic Codes

Structure via **polynomial rings**

- Quotient $\mathbb{F}_q[X]/\langle X^n - 1\rangle$
- Ideals and generator polynomials
- BCH design and cyclic Hamming codes

II. Linear Codes

II. Linear Codes
Linear Code

Linear Encoding

Definition. A **linear encoding** over $\mathbb{F}_q$ is a linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear code

Definition. A **linear** $[n,k,d]_q$ **code** is a k-dimensional subspace

$$C \leq \mathbb{F}_q^n$$

with minimum distance d. The elements of C are called **codewords**.

II. Linear Codes

Generator Matrices

Generator matrix

Definition. A **generator matrix** for an $[n,k]_q$ code is a matrix $\mathbf{G} \in \mathbb{F}_q^{k\times n}$ whose rows form a basis of C. Encoding is the linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{m} \longmapsto \mathbf{mG}.$$

Example (The Hamming Code $[7,4,3]_2$). Consider

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}_{4\times 7}$$

Encode message $\mathbf{m} = (1,0,1,1)$:

$$\mathbf{c} = \mathbf{mG} = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} \cdot \mathbf{G} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}.$$

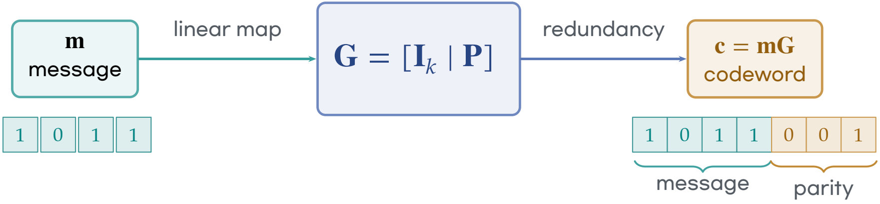
II. Linear Codes
Systematic Form

Systematic Form

Definition. A code is said to be in **systematic form** if it has a generator matrix of the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$$

- The message appears visibly in the first k coordinates.
- The remaining $n - k$ coordinates are parity symbols.
- Systematic form is convenient, but it is not an intrinsic property of the code.



II. Linear Codes
Parity-check Matrices

Parity-check Matrix

A **parity-check matrix** is a matrix $\mathbf{H} \in \mathbb{F}_q^{(n-k)\times n}$ such that

$$C = \ker(\mathbf{H}) = \{\mathbf{c} \in \mathbb{F}_q^n : \mathbf{H}\mathbf{c}^T = 0\}.$$

- If $\mathbf{G}$ generates C, then $\mathbf{G}\mathbf{H}^T = 0$.

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \boxed{\mathbf{H}\mathbf{c}^T} \longrightarrow \mathbf{s} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

II. Linear Codes
Duality

Dual code

Definition. The dual of $C \leq \mathbb{F}_q^n$ is

$$C^\perp = \{\mathbf{y} \in \mathbb{F}_q^n : \langle \mathbf{x}, \mathbf{y} \rangle = 0 \text{ for all } \mathbf{x} \in C\}.$$

Remark (Dimension identity). $\dim C + \dim C^\perp = n$.

II. Linear Codes

Syndromes

Syndrome

Proposition. For a received word $\mathbf{y} = \mathbf{c} + \mathbf{e}$,

$$\mathbf{H}\mathbf{y}^T = \mathbf{H}\mathbf{c}^T + \mathbf{H}\mathbf{e}^T = \mathbf{H}\mathbf{e}^T.$$

- $\mathbf{H}\mathbf{y}^T = 0$ if and only if $\vec{y} \in C$.

II. Linear Codes

Correction and detection

Distance theorem

A code of minimum distance d detects every error of weight at most $d-1$ and corrects every error of weight at most

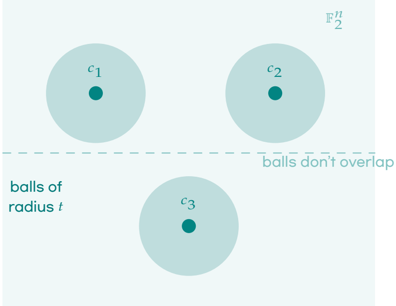
$$t = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

- Detection: no nonzero error of weight $< d$ is a codeword.
- Correction: balls of radius t around distinct codewords are disjoint.

II. Linear Codes

The Packing Picture

Think of $\mathbb{F}_q^n$ as a big space with q^n points. A code places q^k **codewords** in this space, each surrounded by an **error-correction ball** of radius t.



Hamming bound

$q^k \cdot |B_t| \leq q^n$, where $|B_t| = \sum_{i=0}^{t} \binom{n}{i}(q-1)^i$.

Interpretation

- Each codeword "claims" all points within distance t
- These claimed regions cannot overlap
- More codewords ⇒ smaller (or no) balls ⇒ less correction

Perfect code

A code meeting the bound with equality tiles $\mathbb{F}_q^n$ exactly — every

II. Linear Codes
Packing

Hamming bound

If C is an $[n,k,d]_q$ code and $t = \lfloor (d-1)/2 \rfloor$, then

$$q^k \sum_{i=0}^{t} \binom{n}{i} (q-1)^i \leq q^n.$$

Proof.

The balls of radius t around codewords are disjoint subsets of $\mathbb{F}_q^n$. □

Perfect code

Equality means the radius-t balls tile the whole Hamming space.

II. Linear Codes
Singleton bound

Singleton bound

Every $[n,k,d]_q$ code satisfies

$$d \leq n - k + 1.$$

Proof.

Delete $d-1$ coordinates. Distinct codewords must remain distinct, so $q^k \leq q^{n-d+1}$. $\square$

- Codes meeting the bound are called MDS codes.
- Reed-Solomon codes are the standard nonbinary MDS examples.

II. Linear Codes

Gilbert-Varshamov idea

Existence principle

There exist linear codes whose parameters are good because a greedy construction cannot cover the whole space too early.

$$q^k \sum_{i=0}^{d-2} \binom{n-1}{i} (q-1)^i < q^n \quad \Rightarrow \quad \text{an } [n,k,d]_q \text{ code exists.}$$

- This is not a construction one wants to implement.
- It proves that the asymptotic landscape is nontrivial.
- Algebraic families try to approach such existence bounds explicitly.

II. Linear Codes
Griesmer bound

Griesmer bound

For a linear $[n,k,d]_q$ code,

$$n \geq \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil.$$

- This is stronger than Singleton in many small-parameter cases.
- It comes from repeatedly puncturing on the support of a minimum word.
- It is a linear-code bound: linearity is essential.

II. Linear Codes
Puncturing

Puncturing

Puncturing a code at coordinate i means deleting the i-th coordinate from every word.

- Length decreases by one.
- Dimension usually stays the same, but can drop.
- Minimum distance can decrease by at most one.

Use

Puncturing is the basic operation for inductive arguments on code parameters.

II. Linear Codes
Shortening

Shortening

Shortening at coordinate i means first selecting codewords with $c_i = 0$, then puncturing at i.

- Length decreases by one.
- Dimension can decrease by one.
- Minimum distance does not decrease.

Duality

Puncturing and shortening are dual operations:

$$(\text{puncture } C)^{\perp} = \text{shorten } (C^{\perp}).$$

II. Linear Codes

Weight enumerators

Weight enumerator

The weight enumerator is

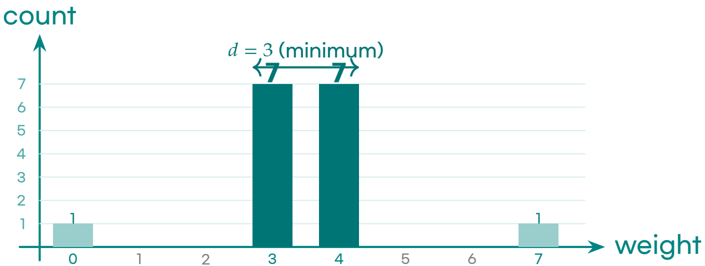
$$W_C(X,Y) = \sum_{c \in C} X^{n-\mathrm{wt}(c)} Y^{\mathrm{wt}(c)}.$$

- It records the distribution of weights.
- The minimum distance is the first nonzero exponent of Y after the constant term.
- It is a coarser invariant than the code but a powerful one.

II. Linear Codes

Weight Distribution of the $[7,4,3]$ Code

The weight enumerator of $\mathrm{Ham}(3,2)$ is $W(X,Y) = X^7 + 7X^4Y^3 + 7X^3Y^4 + Y^7$. This means the 16 codewords have weights:



What we learn

- No codeword has weight 1 or 2 $\Rightarrow d \geq 3$
- The 7 weight-3 words are the Fano plane lines
- Symmetric: C contains the all-ones word

MacWilliams duality

The weight distribution of the dual (simplex) code follows by the MacWilliams transform applied to this bar chart.

II. Linear Codes

MacWilliams identity

MacWilliams identity

For a linear $[n,k]_q$ code,

$$W_{C^\perp}(X,Y) = \frac{1}{|C|} W_C(X+(q-1)Y, X-Y).$$

- The identity is a finite Fourier transform.
- It turns the weight distribution of C into that of $C^\perp$.
- It is one of the central reasons duality is computationally useful.

II. Linear Codes

Self-orthogonality

Self-orthogonal and self-dual

A code is self-orthogonal if $C \subseteq C^{\perp}$ and self-dual if $C = C^{\perp}$.

- Self-dual binary codes have dimension $n/2$.
- Doubly-even self-dual codes have all weights divisible by 4.
- Such codes connect coding theory to lattices and modular forms.

II. Linear Codes
Equivalent codes

Monomial equivalence

Two q-ary linear codes are equivalent if one is obtained from the other by permuting coordinates and multiplying coordinates by nonzero scalars.

- Over $\mathbb{F}_2$, this is just coordinate permutation.
- Equivalent codes have the same parameters and weight enumerator.
- Classification means classification up to this equivalence.

II. Linear Codes
Automorphisms

Automorphism group

$$\mathrm{Aut}(C) = \{\pi \in S_n : \pi(C) = C\}$$

in the binary case.

- A large automorphism group is a sign of hidden geometry.
- Symmetry reduces decoding complexity.
- Hamming and cyclic codes both arise from strong symmetry.

II. Linear Codes

Repetition code

Repetition code

$$C = \{(a, a, \ldots, a) : a \in \mathbb{F}_q\}$$

is an $[n, 1, n]_q$ code.

- It has maximal distance for dimension 1.
- Its rate is $1/n$, so it is inefficient.
- It corrects $\lfloor (n-1)/2 \rfloor$ errors by majority vote when $q = 2$.

II. Linear Codes

Single parity-check code

Parity-check code

$$C = \{x \in \mathbb{F}_q^n : x_1 + \cdots + x_n = 0\}$$

is an $[n, n-1, 2]_q$ code.

- The parity-check matrix is $H = [1\ 1\ \cdots\ 1]$.
- It detects one error but cannot correct one error.
- It is the dual of the repetition code.

II. Linear Codes

The simplex code

Binary simplex code

Let H_r be the matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$. The row space of H_r is the simplex code.

- It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- Every nonzero codeword has the same weight.
- Its dual is the binary Hamming code.

II. Linear Codes
Matroid viewpoint

Vector matroid of a code

A parity-check matrix H defines a matroid on the coordinate set $\{1, \ldots, n\}$ by linear dependence of columns.

- Circuits of the matroid correspond to minimal support codewords.
- The minimum distance is the size of the smallest circuit.
- Projective geometry enters when the columns of H are projective points.

II. Linear Codes

Tanner graphs

Tanner graph

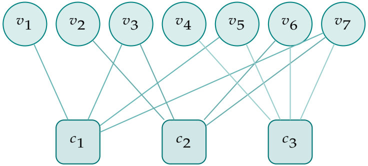
Given a parity-check matrix H, form a bipartite graph with variable nodes $1, \dots, n$ and check nodes equal to the rows of H.

- An edge joins variable j to check i when $H_{ij} \neq 0$.
- Sparse H gives iterative decoding algorithms.
- Classical algebraic codes often have dense checks; LDPC codes emphasize sparsity.

II. Linear Codes

Tanner Graph for the $[7, 4, 3]$ Code

The parity-check matrix H_3 gives a bipartite graph with 7 **variable nodes** and 3 **check nodes**.



How to read the graph

- Variable node v_j = bit j of the codeword
- Check node c_i = parity equation i
- Edge (v_j, c_i) means bit j appears in check i

Each check node says:

$$c_1 \colon v_1 \oplus v_3 \oplus v_5 \oplus v_7 = 0$$

$$c_2 \colon v_2 \oplus v_3 \oplus v_6 \oplus v_7 = 0$$

$$c_3 \colon v_4 \oplus v_5 \oplus v_6 \oplus v_7 = 0$$

LDPC connection

Modern LDPC codes use sparse Tanner graphs — most bits are in only 3-4 checks. This enables fast iterative decoding

II. Linear Codes

Decoding hardness

Syndrome decoding problem

Given H, a syndrome s, and an integer t, decide whether there exists e with

$$He^T = s, \qquad \mathrm{wt}(e) \le t.$$

- This problem is NP-complete in general.
- Code-based cryptography is built around this hardness.
- Algebraic code families are useful because they add trapdoor-like structure.

II. Linear Codes
Perfect codes

Perfect code

A code correcting t errors is perfect if the radius-t balls around codewords partition $\mathbb{F}_q^n$.

$$|C| \sum_{i=0}^{t} \binom{n}{i} (q-1)^i = q^n.$$

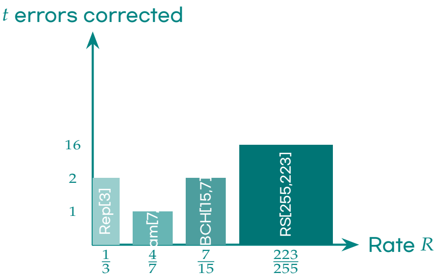
- Perfect codes are extremely rare.
- The binary Hamming codes are perfect one-error-correcting codes.
- The Golay codes are the exceptional perfect codes.

II. Linear Codes

Linear Codes in Practice

Where you use them every day

Application	Code
ECC RAM	Extended Hamming $[72, 64, 4]_2$
RAID-6 disks	Reed-Solomon over $\mathbb{F}_{2^8}$
USB / Ethernet	CRC (cyclic)
QR codes	Reed-Solomon RS$(255, \cdot)$
Satellite TV	Turbo / LDPC codes



Why linear specifically?

- Encoding $m \mapsto mG$ costs $O(nk)$ — fast
- Syndrome Hv^T costs $O(n(n-k))$ — fast
- Parameters can be proved, not just measured

II. Linear Codes

Why linear codes are central

Algebra

- subspaces
- duality
- matrices
- finite fields

Algorithms

- encoding by G
- checking by H
- syndrome decoding
- structured families

II. Linear Codes

The road to Hamming

Question

Can we construct a linear code that corrects one error and wastes no syndrome?

- One error at one of n positions gives n possibilities.
- No error gives one more possibility.
- A parity-check matrix with r rows has 2^r binary syndromes.

$$n + 1 = 2^r$$

is the numerology of binary Hamming codes.

III. Hamming Codes

III. Hamming Codes

Hamming (1950): The Original Problem

Richard Hamming worked at Bell Labs in 1947. The relay computer he used ran **punched-card jobs** overnight — and a single wrong hole would crash the job.

His frustration: "If the machine can detect an error, why can't it fix it?"

Hamming's insight (1950)

Add redundant parity bits so that:

- the syndrome identifies the **exact position** of the error, and
- **no syndrome is wasted** — each possible syndrome maps to exactly one error pattern.

This requires $n+1=2^r$ syndromes for n positions, giving $n=2^r-1$.

Syndrome → action

000 → no error
001 → flip bit 1
010 → flip bit 2
011 → flip bit 3
100 → flip bit 4
101 → flip bit 5
110 → flip bit 6
111 → flip bit 7

III. Hamming Codes

The design problem

Goal

Construct binary linear codes that correct exactly one error and use all possible syndromes.

- A received word has either no error or one error in position i.
- Thus there are $1+n$ elementary cases.
- With r parity checks, there are 2^r syndromes.

$$1+n=2^r \quad \Rightarrow \quad n=2^r-1.$$

III. Hamming Codes

The parity-check matrix

Binary Hamming matrix

For $r \geq 2$, let H_r be the $r \times (2^r - 1)$ matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$.

$$H_3 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- Column labels are nonzero binary syndromes.
- Every nonzero syndrome identifies a unique column.

III. Hamming Codes

Definition

Binary Hamming code

The binary Hamming code of order r is

$$\mathrm{Ham}(r,2) = \ker(H_r) \leq \mathbb{F}_2^{2^r-1}.$$

- Length: $n = 2^r - 1$.
- Redundancy: r.
- Dimension: $k = n - r = 2^r - 1 - r$.

$\mathrm{Ham}(r,2)$ has parameters $[2^r - 1, 2^r - 1 - r, 3]_2$.

III. Hamming Codes

Dimension

Dimension count

The matrix H_r has rank r, hence

$$\dim \ker(H_r) = 2^r - 1 - r.$$

Proof.

The standard basis vectors $e_1, \ldots, e_r$ occur among the nonzero columns of H_r, so the row rank is r. $\square$

Interpretation

The parity checks are independent: no check is redundant.

III. Hamming Codes

Minimum distance

Distance

The binary Hamming code $\mathrm{Ham}(r, 2)$ has minimum distance 3.

Proof.

No column of H_r is zero, so $d \geq 2$. No two columns are equal over $\mathbb{F}_2$, so no two are dependent, hence $d \geq 3$. But $u + v + (u + v) = 0$, so three suitable columns are dependent. $\square$

- Therefore Hamming codes correct one error.
- They detect two errors.

III. Hamming Codes

Perfectness

Hamming codes are perfect

The binary Hamming code satisfies equality in the Hamming bound for $t = 1$.

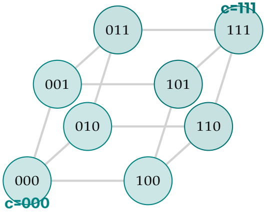
$$2^k(1+n) = 2^{2^r-1-r}(1+2^r-1) = 2^{2^r-1} = 2^n.$$

- Every word of $\mathbb{F}_2^n$ is at distance at most 1 from exactly one codeword.
- The syndrome is not merely a diagnostic: it is a complete decoder.

III. Hamming Codes

Perfect Tiling: A Geometric View

For the $[3,1,3]$ repetition code in $\mathbb{F}_2^3$: the two correction balls of radius 1 **tile the cube exactly** — no overlap, no gap.



Why "perfect"?

- Each ball has 4 elements: 1 codeword + 3 neighbors
- Two balls cover all $2^3 = 8$ points of $\mathbb{F}_2^3$
- $2 \times 4 = 8$ — the Hamming bound holds with equality

For $\mathrm{Ham}(r,2)$

$2^{2^r-1-r} \times (1+2^r-1) = 2^{2^r-1}$ — all of $\mathbb{F}_2^{2^r-1}$ is tiled.

Every received word is within distance 1 of **exactly one** codeword.

Hamming codes are the only linear perfect 1-error-correcting codes (apart from trivial cases).

III. Hamming Codes

Syndrome decoding

Decoder

Given $v \in \mathbb{F}_2^n$, compute $s = H_r v^T$.

- If $s = 0$, accept v.
- If $s \neq 0$, find the unique column $H_i = s$.
- Correct by flipping coordinate i.

Reason

For a single error e_i, the syndrome is $H_r e_i^T = H_i$.

III. Hamming Codes

The Complete Syndrome Table for $[7,4,3]$

With H_3 as written, the syndrome $(s_1, s_2, s_3)^T$ equals column i when bit i is flipped.

Syndrome $(s_1\ s_2\ s_3)$	Error pattern e	Action
$(0,0,0)$	0000000	no error — accept
$(1,0,0)$	1000000	flip bit 1
$(0,1,0)$	0100000	flip bit 2
$(1,1,0)$	0010000	flip bit 3
$(0,0,1)$	0001000	flip bit 4
$(1,0,1)$	0000100	flip bit 5
$(0,1,1)$	0000010	flip bit 6
$(1,1,1)$	0000001	flip bit 7

- All $2^3 = 8$ syndromes are used — confirming **perfectness**
- Decoding is a single table look-up — $O(1)$ hardware

III. Hamming Codes

Binary index convention

Indexed columns

If the i-th column of H_r is the binary expansion of i, then the nonzero syndrome is literally the error position.

$$s = (1,0,1)^T \quad \rightsquigarrow \quad i = 5$$

depending on the chosen bit order.

- This is a notational convention, not an invariant.
- It explains the classical parity-bit positions $1, 2, 4, \ldots$.

III. Hamming Codes
The classical [7,4,3] code

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- There are 7 nonzero syndromes.
- Each syndrome names one error coordinate.
- The kernel has dimension 4.

$$\mathrm{Ham}(3,2) = [7,4,3]_2.$$

III. Hamming Codes

The $[7,4,3]$ Code: Parity Bit Positions

Bits are numbered 1 through 7. Positions that are powers of 2 $(1,2,4)$ are **parity bits**; the rest carry **data**.

p_1	p_2	d_3	p_4	d_5	d_6	d_7
bit 1	bit 2	bit 3	bit 4	bit 5	bit 6	bit 7

Parity check coverage:

Parity bit	Checks bits	Syndrome row
p_1 (bit 1)	1, 3, 5, 7	H row 1: $(1,0,1,0,1,0,1)$
p_2 (bit 2)	2, 3, 6, 7	H row 2: $(0,1,1,0,0,1,1)$
p_4 (bit 4)	4, 5, 6, 7	H row 3: $(0,0,0,1,1,1,1)$

- Each parity bit checks all positions whose index has a 1 in the corresponding binary place
- The 3-bit syndrome tells us the binary address of the flipped bit

III. Hamming Codes

A systematic generator matrix

One systematic form for the $[7,4,3]$ Hamming code is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- The first four coordinates are message symbols.
- The last three are parity symbols.
- This G is equivalent to the kernel of a suitable H.

III. Hamming Codes
Encoding example

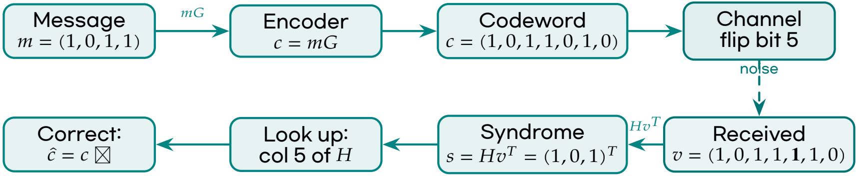
For $m = (1, 0, 1, 1)$ and the systematic matrix G,

$$c = mG = (1, 0, 1, 1, 0, 0, 1).$$

- The parity bits are linear forms in the message bits.
- A single flipped bit changes the syndrome to the corresponding column.
- The correction rule does not require knowing the transmitted message.

III. Hamming Codes

Encoding & Error Correction: The Full Picture



Key numbers:

- Syndrome $s = (1,0,1)^T$ matches column 5 of H_3
- Flip bit 5: $(1,0,1,1,\mathbf{1},1,0) \to (1,0,1,1,0,1,0)$
- Recovered $\hat{c}$ is the original codeword ⊠

The decoder never needs m

The syndrome alone identifies the error. Once we flip the right bit, the first 4 bits of the corrected $\hat{c}$ directly give $\hat{m} = (1,0,1,1)$.

III. Hamming Codes

Projective geometry

Projective interpretation

The nonzero columns of H_r, modulo scalar multiplication, are the points of projective space

$$\mathbb{P}^{r-1}(\mathbb{F}_q).$$

- Over $\mathbb{F}_2$, each nonzero vector is its own projective point.
- Over $\mathbb{F}_q$, scalar multiples define the same projective point.
- Hamming codes are parity checks indexed by projective points.

III. Hamming Codes

The Fano plane

For $r = 3$, the projective space $\mathbb{P}^2(\mathbb{F}_2)$ has 7 points and 7 lines.

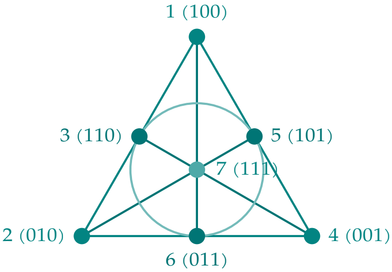
- The 7 columns of H_3 are the points.
- A dependent triple of columns is a line of the Fano plane.
- Weight-3 codewords are exactly these lines.

Geometry

The minimum distance 3 is the statement that the smallest projective line has three points.

III. Hamming Codes

The Fano Plane — Drawn



The 7 lines (each has 3 points):

#	Points	Columns sum to 0
1	$\{1,2,3\}$	$100+010=110$
2	$\{1,4,5\}$	$100+001=101$
3	$\{2,4,6\}$	$010+001=011$
4	$\{1,6,7\}$	$100+011=111$
5	$\{2,5,7\}$	$010+101=111$
6	$\{3,4,7\}$	$110+001=111$
7	$\{3,5,6\}$	$110+101=011$

Why minimum distance $= 3$

Each line is a weight-3 codeword. No point lies alone (weight-1), no pair sums to zero (weight-2). Hence $d = 3$.

III. Hamming Codes

Lines give codewords

In $\mathbb{P}^2(\mathbb{F}_2)$, if a, b, c are the three points on a line, then

$$a + b + c = 0.$$

- The incidence vector of that line is a Hamming codeword.
- The 7 lines give the 7 words of weight 3.
- Their complements give 7 words of weight 4.

$$W_{\mathrm{Ham}(3,2)}(Y) = 1 + 7Y^3 + 7Y^4 + Y^7.$$

III. Hamming Codes

The simplex dual

Simplex code

The row space of H_r is the binary simplex code:

$$\mathrm{Sim}(r,2) = \mathrm{rowspan}(H_r).$$

- It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- Every nonzero word has weight 2^{r-1}.
- It is dual to the Hamming code.

$$\mathrm{Ham}(r,2)^\perp = \mathrm{Sim}(r,2).$$

III. Hamming Codes

MacWilliams check

For $r = 3$, the simplex code has weight enumerator

$$W_{\mathrm{Sim}}(X, Y) = X^7 + 7X^3Y^4.$$

Applying MacWilliams gives

$$W_{\mathrm{Ham}}(X, Y) = X^7 + 7X^4Y^3 + 7X^3Y^4 + Y^7.$$

- The algebraic transform recovers the geometric weight distribution.
- Duality is visible both in matrices and in enumerators.

III. Hamming Codes

Q-ary Hamming codes

Q-ary construction

Choose one nonzero representative from each one-dimensional subspace of $\mathbb{F}_q^r$ and use these vectors as columns of H.

$$n = \frac{q^r - 1}{q - 1}.$$

- The resulting code has parameters

$$\left[\frac{q^r - 1}{q - 1}, \frac{q^r - 1}{q - 1} - r, 3\right]_q.$$

- It is again perfect for one-error correction.

III. Hamming Codes

Why projective representatives

Over $\mathbb{F}_q$, two proportional nonzero columns are linearly dependent.

- To have $d \geq 3$, no two columns may be proportional.
- Therefore columns must be projective points, not raw nonzero vectors.
- Taking all projective points uses the maximum possible number of columns.

Conclusion

Hamming codes are the maximal parity-check configurations with no zero column and no repeated projective point.

III. Hamming Codes

Q-ary perfectness

For the q-ary Hamming code,

$$n = \frac{q^r - 1}{q - 1}, \qquad k = n - r.$$

The radius-one sphere has size

$$1 + n(q-1) = 1 + q^r - 1 = q^r.$$

Hence

$$q^k(1 + n(q-1)) = q^{n-r}q^r = q^n.$$

Perfect again

Every received word lies at distance at most one from a unique codeword.

III. Hamming Codes

The Hamming Family: Parameters at a Glance

r	$n = 2^r - 1$	$k = n - r$	Rate k/n	corrected	Use case
2	3	1	33%	1	tiny
3	**7**	**4**	57%	**1**	**[7,4,3]**
4	15	11	73%	1	medium
5	31	26	84%	1	medium
6	63	57	90%	1	large
$*$	**72**	**64**	89%	**1 (SECDED)**	**ECC RAM**

$*$ Extended Hamming code $[72, 64, 4]_2$: shortened and extended for byte-aligned memory.

- Rate $\to 1$ as $r \to \infty$ — Hamming codes become increasingly efficient
- All correct exactly **one** bit error (single-error-correcting)

III. Hamming Codes

Extended Hamming codes

Extension

Add one overall parity coordinate to force even total weight.

- The binary $[7, 4, 3]$ code becomes an $[8, 4, 4]$ code.
- The new minimum distance is 4.
- It detects three errors and corrects one error.

$$\widehat{\mathrm{Ham}}(3, 2) = [8, 4, 4]_2.$$

III. Hamming Codes

The [8,4,4] code

The extended Hamming code has weight enumerator

$$W(X,Y) = X^8 + 14X^4Y^4 + Y^8.$$

- All codeword weights are divisible by 4.
- The code is self-dual.
- It is the smallest doubly-even self-dual binary code.

Connection

This code is the length-8 seed behind the E_8 lattice construction.

III. Hamming Codes
Automorphisms

Symmetry group

The automorphism group of the binary Hamming code is induced by invertible linear transformations of $\mathbb{F}_2^r$:

$$\mathrm{Aut}(\mathrm{Ham}(r,2)) \cong \mathrm{GL}(r,2).$$

- A matrix in $\mathrm{GL}(r,2)$ permutes the nonzero columns of H_r.
- Hence it permutes coordinates and preserves the kernel.
- The code remembers the projective geometry.

III. Hamming Codes

Tanner graph

- The Tanner graph of H_r has $n = 2^r - 1$ variable nodes and r check nodes.
- It is dense compared with modern LDPC codes.
- But it has an exceptionally simple exact decoder.

Interpretation

For Hamming codes, the syndrome is not iteratively refined; it is read directly as a projective point.

III. Hamming Codes

Failure modes

- A two-error pattern has syndrome equal to the sum of two columns.
- In a binary Hamming matrix, that sum is another nonzero column.
- Therefore a two-error pattern may be miscorrected as a one-error pattern.

Moral

Minimum distance 3 guarantees single-error correction, not reliable detection after attempting correction.

III. Hamming Codes

SECDED

SECDED

Single-error correction, double-error detection is obtained by using the extended Hamming code.

- The extension raises the distance from 3 to 4.
- One error is corrected by the Hamming syndrome.
- Two errors are detected because parity and syndrome disagree in a recognizable way.

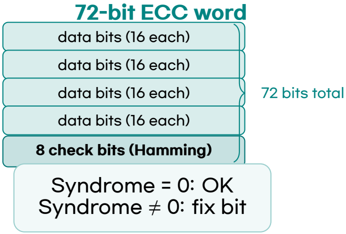
III. Hamming Codes

ECC Memory: Hamming in Your Computer

Error-correcting DRAM (ECC RAM)

Standard server RAM uses the **extended Hamming code** $[72, 64, 4]_2$:

- 64 data bits + 8 check bits per 64-bit memory word
- Corrects any single-bit error automatically
- Detects (but does not correct) any double-bit error
- Rate $= 64/72 \approx 89\%$ — very efficient



Why does it matter?

- Cosmic rays and thermal noise can flip a bit in DRAM roughly once per gigabyte per day
- In a server with 1 TB of RAM, that is ~ 1000 bit flips per day
- ECC corrects each flip silently — the OS

III. Hamming Codes

Shortened Hamming codes

Shortening

Shortened Hamming codes are obtained by fixing selected coordinates to zero and deleting them.

- They are useful when exact lengths $2^r - 1$ are inconvenient.
- The minimum distance remains at least 3.
- Perfectness is usually lost.

Engineering

Many practical memory codes are shortened or extended Hamming codes.

III. Hamming Codes

Hamming codes as BCH codes

Cyclic viewpoint

When $n = 2^r - 1$, the binary Hamming code can be realized as a narrow-sense BCH code of designed distance 3.

- Choose a primitive n-th root $\alpha \in \mathbb{F}_{2^r}$.
- Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- Then the cyclic code generated by $g(X)$ is a Hamming code.

III. Hamming Codes

The [7,4,3] cyclic form

Over $\mathbb{F}_2$,

$$X^7 - 1 = (X+1)(X^3+X+1)(X^3+X^2+1).$$

Taking

$$g(X) = X^3 + X + 1$$

gives a cyclic $[7,4,3]$ Hamming code.

- This is the bridge from projective checks to polynomial ideals.
- The next chapter makes that bridge systematic.

III. Hamming Codes

Classification intuition

- Perfect one-error-correcting linear codes are essentially Hamming codes, apart from trivial cases.
- The parity-check matrix must use every possible nonzero syndrome once.
- Thus the construction is forced by the Hamming bound.

Mathematical taste

Hamming codes are beautiful because optimality, decoding, and geometry are the same statement.

III. Hamming Codes

Exercise: distance

Problem

Let H contain one representative of every point of $\mathbb{P}^{r-1}(\mathbb{F}_q)$. Prove that $\ker H$ has distance exactly 3.

Hint.

- No one or two columns are dependent.
- Any projective line contains three dependent representatives after scaling.
- Translate column dependence into a codeword support.

III. Hamming Codes

Exercise: perfectness

Problem

Verify directly that the q-ary Hamming code meets the Hamming bound with equality for $t = 1$.

$$n = \frac{q^r - 1}{q - 1}, \qquad k = n - r.$$

Computation.

$$q^k(1 + n(q-1)) = q^{n-r}(1 + q^r - 1) = q^n.$$

III. Hamming Codes

Exercise: dual weights

Problem

Show that every nonzero word of the binary simplex code has weight 2^{r-1}.

Hint. For a nonzero row vector $a \in \mathbb{F}_2^r$, the word aH_r evaluates the nonzero linear functional $x \mapsto ax^T$ on every nonzero vector x.

- Exactly half of all vectors of $\mathbb{F}_2^r$ evaluate to 1.
- The zero vector evaluates to 0.

III. Hamming Codes

Summary

Construction

- parity-check columns are projective points
- no repeated projective column
- all syndromes are used

Consequences

- parameters $[2^r - 1, 2^r - 1 - r, 3]_2$
- perfect one-error correction
- dual simplex code

III. Hamming Codes

Transition to cyclic codes

New symmetry

Hamming codes are governed by projective geometry. Cyclic codes are governed by the cyclic shift operator.

- Linear code: subspace of $\mathbb{F}_q^n$.
- Hamming code: subspace defined by projective parity checks.
- Cyclic code: subspace stable under a cyclic permutation.

$$\text{shift symmetry} \quad \Longleftrightarrow \quad \text{ideals in } \mathbb{F}_q[X]/(X^n-1).$$

IV. Cyclic Codes

IV. Cyclic Codes

What Does "Cyclic" Mean?

The core symmetry

A cyclic code is closed under **rotating** the codeword one position to the right.

$c =$ | 1 | 0 | 1 | 1 | 0 | 1 | 0 |

rotate

$c' =$ | 0 | 1 | 0 | 1 | 1 | 0 | 1 |

If $c = (c_0, c_1, \dots, c_{n-1})$ is a codeword, so is

$$c' = (c_{n-1}, c_0, c_1, \dots, c_{n-2}).$$

Applying the shift repeatedly generates all cyclic shifts of c — and all must be in the code.

Why is this useful?

- Shift = multiply by X in the polynomial ring
- Stability under shifts = ideal in R_n
- Ideals are generated by a single generator polynomial $g(X)$
- Encoding = multiply by $g(X)$ — a simple operation

Hardware bonus

Cyclic codes can be encoded and decoded with **shift registers** — fast, cheap circuits. This is why they dominate in hardware applications (CRC, CD, satellite).

IV. Cyclic Codes

Cyclic shift

Cyclic code

A linear code $\mathcal{C} \leq \mathbb{F}_q^n$ is cyclic if

$$(c_0, c_1, \dots, c_{n-1}) \in \mathcal{C} \quad \Rightarrow \quad (c_{n-1}, c_0, \dots, c_{n-2}) \in \mathcal{C}.$$

- This is invariance under a single permutation.
- Since the permutation has order n, cyclic codes are modules over a cyclic group algebra.
- The polynomial model makes this precise.

IV. Cyclic Codes

Polynomial dictionary

Vector to polynomial

Identify

$$(c_0, c_1, \dots, c_{n-1}) \quad \longleftrightarrow \quad c(X) = c_0 + c_1 X + \cdots + c_{n-1} X^{n-1}.$$

- Multiplication by X shifts coefficients.
- The relation $X^n = 1$ performs wrap-around.
- Therefore we work modulo $X^n - 1$.

IV. Cyclic Codes

The quotient ring

Ambient ring

$$R_n = \mathbb{F}_q[X]/(X^n - 1).$$

- Vectors of length n are polynomials of degree $< n$.
- Cyclic shift is multiplication by X in R_n.
- A cyclic code is an $\mathbb{F}_q$-subspace stable under multiplication by X.

Key question

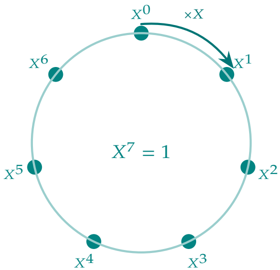
Which subspaces of R_n are stable under multiplication by X?

IV. Cyclic Codes

R_n: The Ring as a Rotation

Think of $R_n = \mathbb{F}_q[X]/(X^n - 1)$ like a **clock with n positions** $0, 1, \dots, n-1$.

Multiplying by X = **rotating one step**.



From integers to polynomials

$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{F}_q[X]/(X^n - 1)$
add n to get 0	$X^n = 1$
$+1$ = shift right	$\times X$ = shift
subgroup = ideal	subspace closed under $\times X$ = ideal

Why this matters

A cyclic code is a subspace of R_n that is closed under rotation. That is exactly an **ideal** — generated by a single divisor $g(X)$ of $X^n - 1$.

IV. Cyclic Codes

Ideals

Cyclic codes are ideals

Under the identification $\mathbb{F}_q^n \cong R_n$, cyclic codes are exactly the ideals of R_n.

Proof.

If C is stable under multiplication by X, then it is stable under multiplication by every polynomial in X. Hence it is an ideal. The converse is immediate. $\square$

Consequence

The classification of cyclic codes is an ideal-theoretic problem.

IV. Cyclic Codes

Principal ideals

Principal generator

Every ideal of R_n is generated by a unique monic divisor $g(X)$ of $X^n - 1$:

$$C = (g(X)).$$

- This follows from the principal ideal property of $\mathbb{F}_q[X]$.
- The generator polynomial is the monic element of least degree in the code.
- The divisibility condition is $g(X) \mid X^n - 1$.

IV. Cyclic Codes
Dimension

Dimension formula

If $\mathcal{C} = (g(X)) \subseteq R_n$ and $\deg g = r$, then

$$\dim_{\mathbb{F}_q} \mathcal{C} = n - r.$$

Proof.

The polynomials

$$g(X), Xg(X), \ldots, X^{n-r-1}g(X)$$

form an $\mathbb{F}_q$-basis of the ideal. $\square$

IV. Cyclic Codes

Generator matrix from shifts

If

$$g(X) = g_0 + g_1 X + \cdots + g_r X^r,$$

then a generator matrix is obtained by shifting the coefficient vector:

$$\begin{bmatrix} g_0 & g_1 & \cdots & g_r & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & \cdots & g_r & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \end{bmatrix}.$$

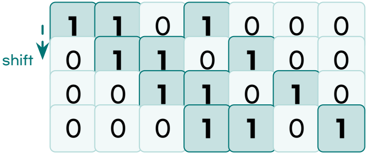
- This matrix is Toeplitz-like.
- It encodes multiplication by $g(X)$.

IV. Cyclic Codes

The Shifted (Toeplitz) Structure of the Generator Matrix

For $g(X) = X^3 + X + 1$ and the $[7,4,3]$ cyclic code, the generator matrix is:

$$G = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$



- Each row is a **one-position cyclic shift** of the row above
- The non-zero diagonals follow the pattern of g's coefficients $(1,1,0,1)$
- This "staircase" pattern is what makes encoding a shift-register operation

IV. Cyclic Codes

Parity-check polynomial

Check polynomial

If $g(X) \mid X^n - 1$, define

$$h(X) = \frac{X^n - 1}{g(X)}.$$

- $\deg h = k$ when C has dimension k.
- Since $g(X)h(X) = 0$ in R_n, multiplication by $h(X)$ checks membership.
- More precisely, $c(X) \in (g)$ implies $c(X)h(X) \equiv 0 \pmod{X^n - 1}$.

IV. Cyclic Codes

Reciprocal polynomials

Reciprocal

For $f(X) = f_0 + \cdots + f_m X^m$ with $f_0 \neq 0$, define

$$f^*(X) = X^m f(X^{-1}).$$

- Parity-check matrices are naturally written using reciprocal polynomials.
- The dual of a cyclic code is cyclic.
- If $C = (g)$ and $h = (X^n - 1)/g$, then $C^\perp$ is generated by a reciprocal of h.

IV. Cyclic Codes

Idempotents

Idempotent generator

When $\gcd(n,q)=1$, each cyclic code has a unique idempotent generator $e(X)$ satisfying

$$e(X)^2 = e(X), \qquad C = (e(X)).$$

- Idempotents decompose the semisimple ring R_n.
- They expose the representation-theoretic structure.
- Generator polynomials are usually better for computation.

IV. Cyclic Codes
Roots

Let α be an n-th root of unity in an extension field.

- A cyclic code generated by $g(X)$ has zeros α^i where $g(\alpha^i) = 0$.
- The defining set is
$$Z(C) = \{i \pmod n : g(\alpha^i) = 0\}.$$
- Root sets translate algebraic divisibility into combinatorics modulo n.

IV. Cyclic Codes

Cyclotomic cosets

Q-cyclotomic coset

Modulo n, the q-cyclotomic coset of i is

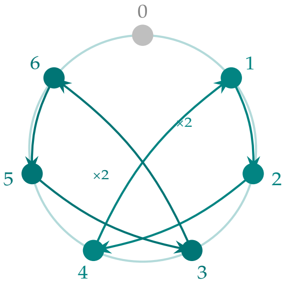
$$C_i = \{i, iq, iq^2, \dots\} \pmod{n}.$$

- If α^i is a root over $\mathbb{F}_q$, all conjugates α^{iq^j} are roots.
- Therefore defining sets are unions of cyclotomic cosets.
- Minimal polynomials correspond to these cosets.

IV. Cyclic Codes

Cyclotomic Cosets for $n = 7$, $q = 2$: Orbits Under $\times 2$

Multiplying by $q = 2$ modulo $n = 7$ partitions $\{0, 1, \ldots, 6\}$ into orbits:



Three cosets:

Coset	Elements	Min. poly
C_0	$\{0\}$	$X+1$
C_1	$\{1,2,4\}$	X^3+X+1
C_3	$\{3,5,6\}$	X^3+X^2+1

This gives the factorization:

$$X^7 - 1 = \underbrace{(X+1)}_{C_0}\underbrace{(X^3+X+1)}_{C_1}\underbrace{(X^3+X^2+1)}_{C_3}$$

Cyclic codes = choose a subset of factors as the generator $g(X)$.

IV. Cyclic Codes

Minimal polynomials

Minimal polynomial

For α^i, the minimal polynomial over $\mathbb{F}_q$ is

$$M_i(X) = \prod_{j \in C_i} (X - \alpha^j).$$

- It lies in $\mathbb{F}_q[X]$.
- It is irreducible.
- Generator polynomials of cyclic codes are products of such $M_i(X)$.

IV. Cyclic Codes

BCH codes

BCH design

A BCH code is a cyclic code whose generator polynomial has a prescribed block of consecutive roots.

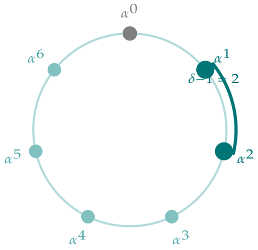
$$g(\alpha^b) = g(\alpha^{b+1}) = \cdots = g(\alpha^{b+\delta-2}) = 0.$$

- The integer δ is the designed distance.
- Consecutive roots force many syndromes to vanish.
- The BCH bound converts this into a distance guarantee.

IV. Cyclic Codes

BCH Design: Consecutive Roots on the Circle

Idea: place n roots of unity on a circle. Prescribe $\delta - 1$ **consecutive** ones as roots of $g(X)$. The BCH bound guarantees $d \geq \delta$.



BCH codes with $n = 7$, $q = 2$:

δ	roots of g	$[n, k]$	d
3	α^1, α^2	$[7, 4]$	3 (Hamming)
5	$\alpha^1, \dots, \alpha^4$	$[7, 1]$	7 (repetition)

BCH Bound

$\delta - 1$ consecutive roots $\Rightarrow d \geq \delta$. More prescribed roots $\Rightarrow$ higher designed distance $\Rightarrow$ lower rate.

Key insight for non-majors: prescribing more roots forces the codeword polynomial to have more zeros, which "spreads" the code and increases the minimum distance — at the cost of fewer codewords (lower rate).

IV. Cyclic Codes

BCH bound

BCH bound

If a cyclic code has $\delta - 1$ consecutive roots

$$\alpha^b, \alpha^{b+1}, \ldots, \alpha^{b+\delta-2},$$

then its minimum distance is at least δ.

Proof.

A codeword of weight $< \delta$ would give a nonzero polynomial with too many consecutive zero syndromes, forcing a Vandermonde determinant to vanish. $\square$

IV. Cyclic Codes

Narrow-sense primitive BCH

Common special case

Let $n = q^m - 1$ and let α be primitive in $\mathbb{F}_{q^m}$. The narrow-sense BCH code of designed distance δ has roots

$$\alpha, \alpha^2, \dots, \alpha^{\delta-1}.$$

- Primitive: the length is the full multiplicative order.
- Narrow-sense: the first root is α.
- Hamming codes occur when $\delta = 3$ in the binary case.

IV. Cyclic Codes

Cyclic Hamming code

Let $n = 2^r - 1$ and let $\alpha \in \mathbb{F}_{2^r}$ be primitive.

- Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- Then $\deg g = r$.
- The cyclic code (g) has parameters

$$[2^r - 1, 2^r - 1 - r, 3]_2.$$

Identification

This is the binary Hamming code in cyclic coordinates.

IV. Cyclic Codes

The case n equals 7

Over $\mathbb{F}_2$,

$$X^7 - 1 = (X+1)(X^3+X+1)(X^3+X^2+1).$$

- A primitive 7-th root has minimal polynomial X^3+X+1 or X^3+X^2+1.
- Either cubic gives a cyclic $[7,4,3]$ Hamming code.
- The two choices are equivalent under coordinate reversal.

IV. Cyclic Codes

Encoding by multiplication

For a cyclic code (g) with $\deg g = r$, non-systematic encoding is

$$m(X) \longmapsto c(X) = m(X)g(X), \qquad \deg m < n - r.$$

- The message polynomial has $k = n - r$ coefficients.
- The product automatically lies in the ideal (g).
- Hardware can implement this multiplication by shift registers.

IV. Cyclic Codes

Systematic cyclic encoding

Let $\deg g = r$. To encode a message $m(X)$ with $\deg m < k$:

$$X^r m(X) = q(X)g(X) + p(X), \qquad \deg p < r.$$

Then set

$$c(X) = X^r m(X) - p(X).$$

- Since $c(X)$ is divisible by $g(X)$, it is a codeword.
- The high-degree coefficients contain the message.
- The low-degree polynomial $p(X)$ supplies parity.

IV. Cyclic Codes

Systematic Encoding: Step-by-Step Example

Encode message $m(X) = X^3 + 1$ (i.e. $m = (1,0,0,1)$) using $g(X) = X^3 + X + 1$.

Step 1. Form $X^3 m(X) = X^6 + X^3$

Step 2. Divide: $X^6 + X^3 = (X^3 + X^2 + X + 1) \cdot g(X) + (X^2 + X)$

Step 3. Remainder $p(X) = X^2 + X$ (the parity part)

Step 4. Codeword: $c(X) = X^3 m(X) - p(X) = X^6 + X^3 + X^2 + X$

Check: $c(X) \div g(X) = (X^3 + X^2 + X + 1)$ — remainder 0 ⊠

Codeword in vector form:

$$c = (0,1,1,1,0,0,1) \in \mathbb{F}_2^7$$

- Bits 4-7 (positions $X^3, \dots, X^6$): the message $(1,0,0,1)$
- Bits 1-3: the parity $(0,1,1)$ from $p(X) = X^2 + X$

Why this works

Since $c(X) = X^3 m(X) - p(X)$ and $p(X) \equiv X^3 m(X) \pmod{g}$, we get $c(X) \equiv 0 \pmod{g}$. So c is a codeword.

IV. Cyclic Codes

Syndrome polynomial

For received $v(X) = c(X) + e(X)$, compute the remainder

$$s(X) = v(X) \bmod g(X).$$

- Since $c(X)$ is divisible by $g(X)$, $s(X) = e(X) \bmod g(X)$.
- This is the polynomial analogue of $Hv^T = He^T$.
- BCH decoding refines this by evaluating at roots of g.

IV. Cyclic Codes

Root syndromes

If α^i is a root of $g(X)$, then every codeword satisfies

$$c(\alpha^i) = 0.$$

For $v = c + e$,

$$v(\alpha^i) = e(\alpha^i).$$

- The numbers $S_i = v(\alpha^i)$ are syndromes.
- Decoding asks for the error locations and values from these sums.
- This is a finite-field moment problem.

IV. Cyclic Codes

Error locator polynomial

For error locations $j_1, \dots, j_t$, define

$$\Lambda(Z) = \prod_{\ell=1}^{t} (1 - \alpha^{j_\ell} Z).$$

- The roots of Λ identify error locations.
- BCH and Reed-Solomon decoding compute Λ from syndromes.
- Berlekamp-Massey is an efficient algorithm for this recurrence problem.

IV. Cyclic Codes

LFSR viewpoint

Linear feedback shift register

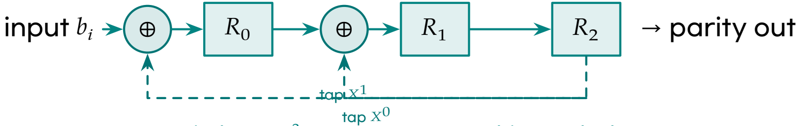
Division by $g(X)$ can be implemented with a shift register whose feedback taps are the nonzero coefficients of $g(X)$.

- Memory cost is $\deg g = n - k$ registers.
- Input is processed one symbol per clock.
- This is why cyclic codes became central in hardware error control.

IV. Cyclic Codes

The LFSR Circuit for $g(X) = X^3 + X + 1$

For the $[7,4,3]$ cyclic Hamming code, $g(X) = X^3 + X + 1$. The feedback taps are at X^0 and X^1.



Computes remainder of $X^3 \cdot m(X) \div g(X)$ — one bit per clock

Encoding (4 clock cycles):

1. Feed in the 4 message bits m_3, m_2, m_1, m_0
2. The 3 registers accumulate the parity
3. Read off (R_0, R_1, R_2) — these are the 3 parity bits

Key advantage:

- Only $r = n-k$ flip-flops needed
- One XOR gate per feedback tap
- Works at wire speed — no computation delay
- Used in every USB, Ethernet, and SATA controller

IV. Cyclic Codes

Burst errors

Burst

A burst of length ℓ is an error whose nonzero positions lie in ℓ consecutive coordinates.

- Polynomial remainders are well suited for burst detection.
- CRC polynomials are chosen to detect common burst patterns.
- This is detection rather than correction in many communication protocols.

IV. Cyclic Codes
CRC codes

CRC uses a generator polynomial $g(X)$ and transmits a multiple of $g(X)$.

- Receiver divides by $g(X)$.
- Nonzero remainder means an error was detected.
- The method is a cyclic-code membership test.

Mathematical core

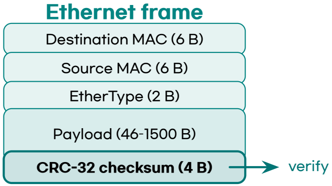
CRC is not a different theory: it is cyclic codes used primarily for error detection.

IV. Cyclic Codes

CRC in Your Daily Life

Every data transfer uses CRC.

Standard	Polynomial	Detects
Ethernet (CRC-32)	$x^{32} + \cdots$	≤ 32-bit bursts
USB 2.0	CRC-16	≤ 16-bit bursts
SATA/NVMe	CRC-32	storage errors
PNG / ZIP	CRC-32	file corruption
Bluetooth	CRC-24	packet errors



CRC detects $\approx 99.9998\%$ of random errors. It does not correct them — for correction, add an outer code (e.g. Reed-Solomon).

How it works (in one sentence):

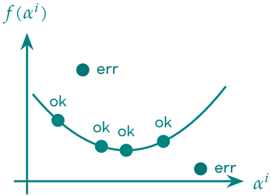
- Sender appends $r = \deg g$ bits so the extended packet is divisible by $g(X)$
- Receiver divides: nonzero remainder $\Rightarrow$ error detected

IV. Cyclic Codes

Reed-Solomon: The Core Idea

The polynomial evaluation idea

A degree-$(k-1)$ polynomial over $\mathbb{F}_q$ is determined by k of its values. If you evaluate it at $n > k$ points and some values are corrupted, you can still recover the polynomial — because any k correct values suffice.



Reed-Solomon $\mathrm{RS}(n,k)$

- Message: polynomial f of degree $< k$
- Codeword: $(f(\alpha^0),\dots,f(\alpha^{n-1}))$
- Minimum distance: $d = n - k + 1$ **(MDS)**
- Corrects up to $\lfloor (n-k)/2 \rfloor$ errors

In practice:

- CD, DVD, Blu-ray: RS$(255, 251)$ per 8-bit symbol
- QR codes: up to RS$(255, 223)$
- Deep space (Voyager): RS$(255, 223)$, corrects up to 16

IV. Cyclic Codes

Reed-Solomon codes

Reed-Solomon

Over $\mathbb{F}_q$, Reed-Solomon codes evaluate polynomials of degree $< k$ at n distinct field points.

- In a suitable form, they are cyclic codes.
- They meet the Singleton bound: $d = n - k + 1$.
- They correct symbol errors, which is why they are strong against bursts.

IV. Cyclic Codes

BCH versus Reed-Solomon

BCH

- often binary
- roots in an extension field
- designed distance
- strong for bit errors

Reed-Solomon

- nonbinary
- evaluation over $\mathbb{F}_q$
- MDS property
- strong for symbol bursts

IV. Cyclic Codes

Dual cyclic codes

Dual cyclicity

The dual of a cyclic code is cyclic.

Proof.

If c is orthogonal to every word of $\mathcal{C}$, then a cyclic shift of c is also orthogonal to every cyclic shift of every word of $\mathcal{C}$. Since $\mathcal{C}$ is shift-invariant, the shifted c lies in $\mathcal{C}^{\perp}$. □

- Generator polynomials of duals are expressed using reciprocal polynomials.
- This is another sign that cyclic codes form a closed algebraic world.

IV. Cyclic Codes

Repeated-root caution

Assumption often made

Many clean statements assume $\gcd(n, q) = 1$.

- Then $X^n - 1$ has no repeated roots.
- The quotient ring is semisimple.
- If $\gcd(n, q) \neq 1$, repeated-root cyclic codes require extra care.

Moral

Always check separability before using root-set arguments naively.

IV. Cyclic Codes

Quasi-cyclic hint

Quasi-cyclic code

A code is quasi-cyclic of index ℓ if it is invariant under cyclic shift by ℓ positions.

- Generator matrices are built from circulant blocks.
- Circulant blocks correspond to polynomials modulo $X^m - 1$.
- This structure is important in code-based cryptography.

IV. Cyclic Codes

Algebraic hierarchy

$$\begin{array}{c} \text{linear code} \\ \Downarrow \text{ add cyclic shift} \\ \text{cyclic code} \\ \Downarrow \text{ prescribe roots} \\ \text{BCH code} \\ \Downarrow \text{ evaluation/MDS structure} \\ \text{Reed-Solomon code} \end{array}$$

Perspective

Each additional algebraic structure buys stronger theorems and faster algorithms.

IV. Cyclic Codes

Cyclic codes as modules

- A linear code is an $\mathbb{F}_q$-subspace of $\mathbb{F}_q^n$.
- A cyclic code is an $\mathbb{F}_q[X]$-submodule, where X acts by shift.
- The relation $X^n = 1$ makes the module a module over R_n.

$$\text{cyclic code} = \text{submodule of } R_n = \text{ideal of } R_n.$$

Mathematician's slogan

Shift-invariant linear algebra is ideal theory.

IV. Cyclic Codes

Exercise: ideal proof

Problem

Prove that if an $\mathbb{F}_q$-subspace $C \subseteq R_n$ is closed under multiplication by X, then it is an ideal.

Hint.

- Closure under X gives closure under X^j for every $j \geq 0$.
- By linearity, closure under every polynomial follows.
- Reduce modulo $X^n - 1$.

IV. Cyclic Codes

Exercise: dimension

Problem

Let $g(X) \mid X^n - 1$ and $\deg g = r$. Show that (g) has dimension $n - r$ as an $\mathbb{F}_q$-space.

Hint.

$$a(X)g(X), \qquad \deg a < n - r$$

are all distinct modulo $X^n - 1$.

- Count the possible coefficients of $a(X)$.
- Use the degree bound to prove injectivity.

IV. Cyclic Codes

Exercise: cyclic Hamming

Problem

For $X^7 - 1 = (X+1)(X^3+X+1)(X^3+X^2+1)$ over $\mathbb{F}_2$, show that the code generated by $g = X^3 + X + 1$ has parameters $[7, 4, 3]$.

Hint.

- Dimension is $7 - \deg g = 4$.
- The BCH bound with roots α, α^2 gives $d \geq 3$.
- The Singleton bound excludes $d > 4$, and a direct word gives $d = 3$.

IV. Cyclic Codes

Summary of cyclic codes

Algebra

- $R_n = \mathbb{F}_q[X]/(X^n - 1)$
- cyclic codes are ideals
- generator polynomial g
- root sets and cosets

Algorithms

- encoding by multiplication
- parity by remainders
- BCH decoding from syndromes
- LFSR implementation

IV. Cyclic Codes

Final perspective

Linear codes turn error correction into finite-dimensional algebra.
Hamming codes show how perfect decoding is finite geometry.
Cyclic codes show how shift symmetry becomes polynomial ideal theory.

Coding theory is the art of adding structure to redundancy.

Thank You

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